

Notes on

This is a collection of progress notes on the architecture, training and challenges of the Network used for this project, which can be used as reference when writing the final report:

Data Preparation

There is an encountered difficulty in the data set due to varying wafer map sizes:

- 632 different format sizes for the wafer maps, differing in aspect ratio and varying from 25x25 for the smallest to 300x300 for the biggest.
- To Tackle this an additional step of data cleaning/preparation was taken
- To preserve information and features, simple rescaling is not enough
- since the input map is initially an integermap(0 for background, 1 for no marking, 2 for defect marking) there is a need for rescaling without averaging over neighbouring pixels, so a nearest neighbor rescaling is taken. (**maybe**)
- To avoid making features less recognizable by warping, before rescaling padding is added(pixels of 0 for background)
- To not make the Network too large but to also not lose too much information and finer details on larger wafers a scale of 64x64, close to the overall average size, is chosen.